

Shahriar Sakib

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 Portfolio  github.com/sakibisactive  LinkedIn

PROFILE

Final-year Computer Science & Engineering student at BRAC University with hands-on experience across full-stack web development, applied machine learning, database systems, and IoT. Currently conducting undergraduate thesis research on bio-inspired swarm routing algorithms. Passionate about writing reliable, production-ready software and building systems that solve real-world problems.

EDUCATION

B.Sc. in Computer Science & Engineering	2022 – 2026
BRAC University, Dhaka · CGPA: 3.45 / 4.00	
Higher Secondary Certificate, Science	2020
Dhaka City College, Dhaka · GPA: 5.00 / 5.00	
Secondary School Certificate, Science	2018
Govt. Laboratory High School, Dhaka · GPA: 5.00 / 5.00	

TECHNICAL SKILLS

Languages	Python, JavaScript, PHP, SQL, C, C++
ML / Data	NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow/Keras
Deep Learning	CNN, RNN, LSTM, GRU, Transformer, ViT · DenseNet, ResNet, MobileNet
Web / Full-Stack	React.js, Node.js, Express.js, Socket.IO, Next.js, Laravel, PHP, REST APIs
Databases	MySQL, MongoDB Atlas, Redis (Upstash)
Tools & DevOps	Git, GitHub, Vercel, Jupyter Notebook, VS Code, Overleaf, SUMO
Interests	Full-Stack Development, Machine Learning, NLP, Computer Vision, IoT, Swarm AI

UNDERGRADUATE THESIS

E³-Hybrid Swarm Routing for EV Emergency Coordination	2025 – 2026
Supervisor: Dr. Md. Khalilur Rahman, BRAC University · Status: In Progress	
- Proposes E³-Hybrid, a novel algorithm combining ACO, PSO, and BCO for decentralised multi-agent EV routing under emergency conditions in urban environments. - Runs 1,050 simulation batches (7 algorithms × 6 stress scenarios × 25 seeds) on a SUMO-based Midtown Manhattan network with full statistical analysis and IEEE-format visualisations.	

PROJECTS

SoulSync — Partner Matching Web Platform Live GitHub 2025

MERN Stack · React 19 · TypeScript · Redux Toolkit + RTK Query · Socket.IO · Cloudinary · JWT Auth · Tailwind CSS

- Built a 5-dimensional weighted matching engine (Personality 35%, Interests 25%, Lifestyle 20%, Age 10%, Location 10%) with real-time Socket.IO chat, typing indicators, read receipts, and multi-photo upload via Cloudinary.
- JWT access/refresh tokens, bcrypt hashing, HTTP-only cookies, RBAC (Guest/User/Admin), and a full admin dashboard with user analytics, report queues, and moderation tools.

ConnectHub — Real-Time Messaging Platform Live GitHub 2026

MongoDB Atlas, Express.js, React.js, Node.js · Socket.IO · Redis (Upstash) · JWT Auth

- Built a production-grade real-time chat app with Socket.IO WebSocket messaging and Upstash Redis caching, deployed live on Vercel.
- Implemented JWT authentication, encrypted WebSocket rooms, persistent message history with MongoDB Atlas, and a responsive UI.

E-Commerce Platform Live GitHub 2025

MongoDB, Express.js, React.js, Node.js (MERN) · JWT Auth · Tailwind CSS

- Built a full-stack e-commerce application with dynamic product catalogs, JWT authentication, and cart management.
- Deployed live on Vercel with RESTful API backend and persistent MongoDB order storage.

Doctor Handwriting Recognition GitHub 2026

Python, TensorFlow/Keras · DenseNet121, MobileNetV3, ResNet50, ViT-B16, CNN-Transformer

- Benchmarked five DL architectures (DenseNet121, MobileNetV3, ResNet50, ViT-B16, CNN-Transformer) for prescription handwriting classification; built full pipeline from preprocessing to cross-model evaluation.

News Headline Text Classification GitHub 2025

Python, Scikit-learn, TensorFlow · SimpleRNN, LSTM, GRU, Logistic Regression, DNN

- Evaluated 24 NLP pipeline configurations (TF-IDF, embeddings, RNN/LSTM/GRU); achieved best macro F1 of 0.9276 with Bidirectional GRU, documented in an IEEE-format report.

A.Q.U.A.: Aquarium Automation System 2025

Arduino Uno R3, ESP8266, IoT Sensors, Embedded C

- Developed an IoT-enabled automation system with scheduled feeding, water quality monitoring, and real-time WiFi telemetry.

LANGUAGES

Bangla	Native
English	Proficient (Academic & Professional)